

Technical Document #149: Blockchain - Comprehensive Overview

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Tokenomics studies the economic systems of cryptocurrency tokens. It covers supply mechanisms, incentive structures, and value capture.

Proof of Stake (PoS) selects validators based on their stake in the network. It is more energy-efficient than Proof of Work.

Non-fungible tokens (NFTs) represent unique digital assets on the blockchain. They have transformed digital ownership and art markets.

Blockchain is a distributed ledger technology that records transactions across many computers. It ensures transparency and immutability.

Consensus mechanisms ensure all nodes in a network agree on the state of the blockchain. They are essential for maintaining network integrity.

Autonomous vehicles use a combination of sensors, AI, and mapping to navigate without human intervention. Self-driving technology is rapidly advancing.

Sustainability and environmental responsibility are becoming key considerations in technology design. Green computing aims to reduce the environmental impact of technology.

Quantum computing promises to solve problems that are intractable for classical computers. It has potential applications in cryptography, drug discovery, and optimization.

Key Takeaways:

- Non-fungible tokens (NFTs) represent unique digital assets on the blockchain.
- Zero-knowledge proofs allow one party to prove knowledge of information without revealing the information itself.
- Smart contracts are self-executing contracts with terms directly written into code.